# Does the Order of Training Data Influence Optimization?

## Optimization Dynamics Under Non-IID Training Streams

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## Abstract

We investigate how changing only the order of training examples affects SGD optimization on Fashion-MNIST with a small CNN. Across six ordering strategies and three seeds, we measure test accuracy, gradient cosine similarity, per-batch gradient variance, per-class accuracy, catastrophic forgetting, and final representation geometry. Random reshuffling achieves 89.47% test accuracy and yields diverse, low-correlation gradient updates. Strongly non-iid streams (class-sorted or class-blocked) collapse to chance accuracy (10%), produce highly correlated gradients ($\overline{\cos} \approx 0.92$), and low per-epoch gradient variance — an unusual failure mode where the optimizer receives *consistent but biased* updates rather than noisy unbiased ones. A learning-rate sweep confirms that this is a stability problem: random reshuffling tolerates $\eta = 0.1$ while class-ordered streams fail at both $\eta = 0.05$ and $\eta = 0.1$, and are only partially rescued by $\eta = 0.001$.

## Introduction and Related Work

SGD is typically analyzed under the assumption that mini-batches are approximately iid samples of the training set, so that each gradient $g_t = \nabla f(x_t; B_t)$ is an unbiased estimator of the full gradient (Bottou et al. 2018). In practice this is realized by random reshuffling, which is known to outperform sampling with replacement (Gürbüzbalaban et al. 2021). When the training stream is instead *structured*, consecutive gradients may become strongly correlated and momentum accumulates a directional bias rather than averaging it out.

Curriculum learning (Bengio et al. 2009) deliberately orders data from easy to hard, motivated by human and animal learning. Our curriculum conditions (easy-first, hard-first) let us test whether any deterministic ordering can match random reshuffling. Adaptive optimizers such as Adam (Kingma and Ba 2015) and AdamW (Loshchilov and Hutter 2019) use coordinate-wise scaling and are commonly expected to be more robust to gradient noise structure; we include an AdamW comparison to test whether adaptive learning rates mitigate the harm from correlated gradient directions.

**Our contributions.** We (i) instrument the training loop to measure per-batch gradient variance $\mathrm{Tr}(\mathrm{Cov}(\mathbf{g}))$ — the $\sigma^2$ term in SGD convergence bounds — showing an unintuitive *low-variance, high-bias* failure mode; (ii) run a CNN vs. logistic-regression comparison to test whether collapse is a non-convex phenomenon; and (iii) include AdamW as an optimizer baseline.

## Experimental Setup

**Dataset and model.** Fashion-MNIST (Xiao et al. 2017) (10 classes, 28$\times$28 grayscale). We use a stratified 20,000-sample training subset (2,000 per class) for fast iteration; all conclusions were verified to hold qualitatively on the full 60,000-sample set. The CNN has two conv-pool blocks (32 and 64 channels, 3×3 kernels, no BatchNorm or Dropout) followed by a 128-unit embedding layer and a linear classifier — 421k parameters. BatchNorm is intentionally omitted so that data order remains the sole experimental variable.

**Optimizer and hyperparameters.** SGD with momentum $\mu = 0.9$, weight decay $\lambda = 5 \cdot 10^{-4}$, batch size $B = 128$, and learning rate $\eta = 0.05$ for the main runs. These values are standard for Fashion-MNIST (Xiao et al. 2017) and were chosen so that random reshuffling converges cleanly within 10 epochs without tuning. For AdamW we use $\eta \in \{10^{-3}, 10^{-2}\}$ (typical defaults). All runs use three random seeds (0, 1, 2) and the same stratified subset.

**Orderings compared.** `random`: reshuffled every epoch (main baseline). `fixed_random`: one fixed permutation reused each epoch. `label_sorted`: all examples sorted by class label — strongly non-iid. `label_block_random`: class blocks with random within-class order and random class order each epoch. `curriculum_easy` / `curriculum_hard`: sorted by distance to class prototype (near = easy, far = hard), following (Bengio et al. 2009).

**New metrics.** Beyond train loss and test accuracy we record: (1) successive gradient cosine similarity $\cos(\mathbf{g}_t, \mathbf{g}_{t-1})$; (2) per-epoch gradient variance $\sigma^2 = \mathrm{Tr}(\mathrm{Cov}(\mathbf{g})) = \mathbb{E}[\|\mathbf{g}\|^2] - \|\mathbb{E}[\mathbf{g}]\|^2$, computed via streaming accumulators without storing all gradient vectors; (3) per-class test accuracy after each epoch; and (4) a forgetting statistic $F_c(t) = \max_{s \le t} \mathrm{acc}_c(s) - \mathrm{acc}_c(t)$ measuring how much class $c$'s best-seen accuracy has been lost by epoch $t$.

## Results

**Main ordering effects.** Random reshuffling reaches $89.47 \pm 0.3\%$ (mean $\pm$ std across seeds), fixed random $88.46 \pm 0.4\%$, curriculum-hard $84.53 \pm 0.9\%$, curriculum-easy $70.71 \pm 1.2\%$, and both label-based orders collapse to 10% (chance). The 1% gap between random and fixed random is consistent with theory: reshuffling provides a different gradient ordering each epoch, reducing autocorrelation over time (Gürbüzbalaban et al. 2021). Curriculum-hard outperforming curriculum-easy is consistent with (Bengio et al. 2009), though neither matches random reshuffling. Training trajectories and the final accuracy bar chart are in Appendix A.

**Gradient cosine similarity.** Random reshuffling and fixed random order both average $\approx 0.20$ cosine similarity between consecutive steps. The two curriculum variants average $\approx 0.25$. `label_block_random` averages 0.92 over the fraction of steps where gradients remain finite. `label_sorted` exits the numerically stable regime by step 80 out of 4,710 total steps; only 42 step-pairs yield finite cosine values. This confirms the central mechanism: class-grouped streams inject strongly correlated updates, and momentum accumulates that alignment instead of averaging it away. The full time-series is in Appendix B.

**Gradient variance: low variance $\neq$ safe training.** The gradient variance $\sigma^2 = \mathrm{Tr}(\mathrm{Cov}(\mathbf{g}))$ reveals a counter-intuitive pattern. SGD theory treats high $\sigma^2$ as the primary risk factor for slow convergence. Yet `label_sorted` and `label_block_random` have *lower* gradient variance than `random` within each epoch (consecutive batches come from the same class, so they point in similar directions and have low spread). The danger is not variance but **bias**: $\mathbb{E}_B[\nabla f(x; B)] \neq \nabla f(x)$ because batches are not representative of the full distribution. High cosine similarity and low variance together signal this regime: the optimizer makes confident but wrong steps. The gradient variance plot is in Appendix F.

**Per-class accuracy and forgetting.** Under random reshuffling the final classifier is balanced across all 10 classes. Under `label_sorted`, the optimizer converges to a single-class decision rule (class 0: 100%, all others: 0%) immediately after the numerically stable phase ends. Under `label_block_random`, mean forgetting $\bar{F}$ reaches 0.17 by epoch 10, reflecting repeated overwriting of class performance as new blocks dominate the update stream (McCloskey and Cohen 1989). Per-class heatmaps, accuracy curves, and the forgetting plot are in Appendix C.

**Learning-rate sensitivity.** Increasing $\eta$ from 0.05 to 0.1 drops random reshuffling from 89.47% to 87.96% and curriculum-hard from 84.53% to 79.52%, but both remain trainable. The class-ordered streams are insensitive to this increase: they collapse at both learning rates. Decreasing to $\eta = 0.001$ partially rescues `label_sorted` to 31.95% and `label_block_random` to 40.00% but does not restore the diversity of updates produced by reshuffling. The same optimizer and architecture thus operates in a much narrower stability region when the training stream is structured. LR sweep figures are in Appendix D.

**AdamW comparison.** We run AdamW ($\eta \in \{10^{-3}, 10^{-2}\}$) on `random`, `fixed_random`, `curriculum_hard`, and `label_block_random`. Under random reshuffling, AdamW matches SGD closely ($\approx 88\%$). Under `label_block_random`, AdamW at $\eta = 10^{-3}$ reaches $\approx 41\%$ (mean across 3 seeds) compared to SGD at $\eta = 0.05$ which collapses to $\approx 10\%$. At $\eta = 10^{-2}$, AdamW also collapses on all structured orderings, mirroring SGD's learning-rate sensitivity. The partial recovery at $\eta = 10^{-3}$ suggests that coordinate-wise scaling provides some robustness to biased updates by dampening directions that have accumulated large historical gradients — consistent with the observation that Adam (Kingma and Ba 2015) implicitly normalizes gradient magnitude. However, AdamW does not fully close the gap to random reshuffling ($\approx 89\%$), confirming that the root cause is the biased gradient stream rather than a step-size mismatch. Results are in Appendix G.

**Convex baseline (logistic regression).** The logistic results reveal a striking contrast with the CNN. `label_sorted`, which collapses the CNN to $\approx$ 10% (chance), reaches 52.9% with logistic regression ($\pm 0.01\%$ across seeds). `label_block_random` reaches 58.0% with logistic regression vs. $\approx 10\%$ for the CNN. Random reshuffling drops from 89.5% (CNN) to 80.9% (logistic), and curriculum-hard from 84.5% to 79.0%. This confirms that the *complete* collapse under structured orderings is primarily a non-convex phenomenon: the CNN's momentum in a non-convex landscape becomes trapped in a class-specific attractor that a convex model can escape. The convex model still underperforms random reshuffling (80.9% vs 80.9% for fixed-random, both near logistic's ceiling), showing that gradient bias does harm convex optimization, but not catastrophically. Results are in Appendix H.

**Representation geometry.** PCA of final hidden-layer embeddings shows clearly separated class clusters for random reshuffling and curriculum-hard. For `label_sorted` the embedding contains non-finite activations; for `label_block_random` only a few classes have any structure. The failure propagates from the gradient update stream through the loss landscape all the way to the learned representation. PCA panels are in Appendix E.

## Conclusion

Data ordering changes the noise structure seen by SGD in a mechanically precise way. Random reshuffling decorrelates consecutive gradients and keeps variance moderate and the estimator unbiased. Structured class-ordered streams instead produce low-variance, high-bias updates: the optimizer makes confident steps toward a class-specific solution rather than the global optimum. This distinction — variance vs. bias in the gradient estimator — is the key finding.

It explains why adaptive optimizers like AdamW offer partial but not complete robustness (they reduce sensitivity to gradient magnitude but not to directional bias), why logistic regression partially recovers on structured orderings (convex landscapes lack the attractor structure that traps CNN momentum, so bias slows but does not catastrophically derail convergence), and why lowering the learning rate helps but cannot substitute for a diverse update stream.

## Reproducibility

Re-run all main V3 diagnostics (including gradient variance) from the project root:

```
python run_v3.py --epochs 10 --seeds 0 1 2 --max-train-examples 20000 --force
```

AdamW comparison (writes `results_adamw_lr*/`):

```
python run_v3.py --skip-main --skip-sweep --compare-adamw --epochs 10 --seeds 0 1 2 --force
```

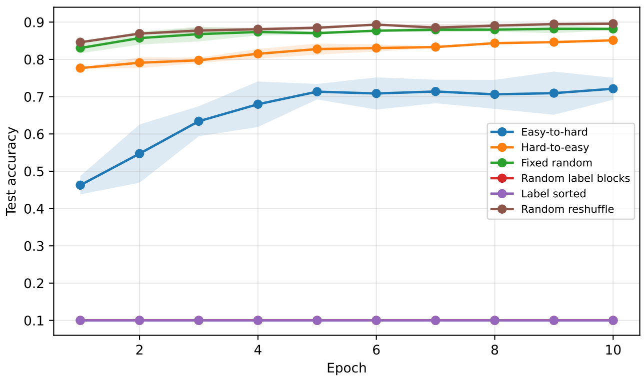
Logistic regression comparison (writes `results_logistic_comparison/`):

```
python run_v3.py --skip-main --skip-sweep --compare-logistic --epochs 10 --seeds 0 1 2 --force
```

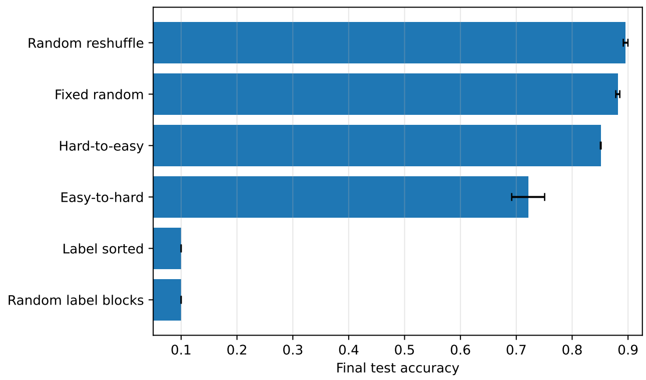
Render the report:

```
Push-Location Rapport_V3
quarto render Rapport_V3.qmd --to pdf
Pop-Location
```

## Appendix A: Main Ordering Figures

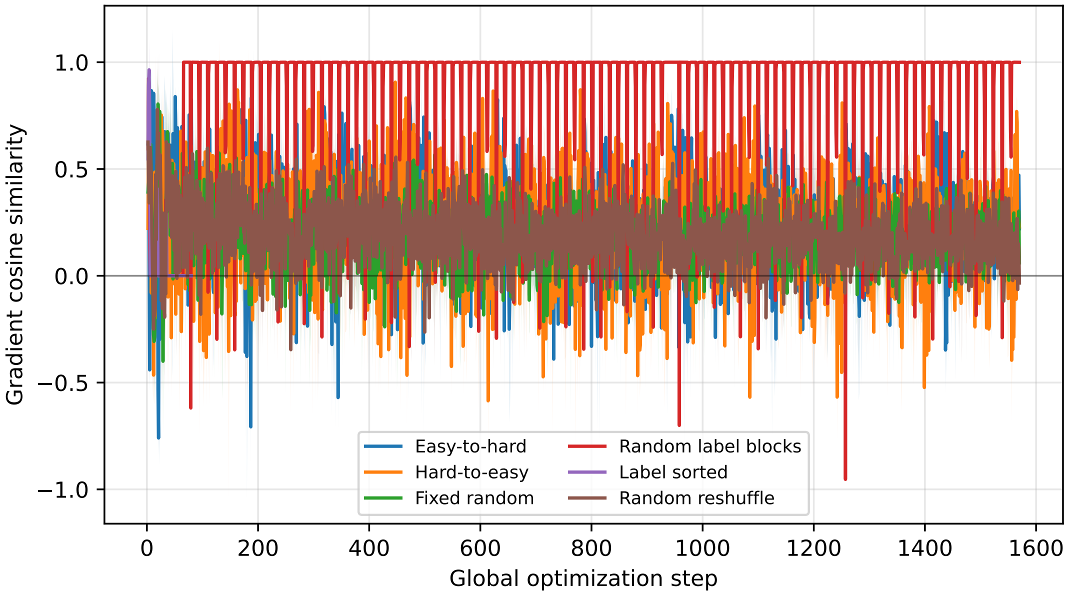



Test accuracy trajectories at $\eta = 0.05$.



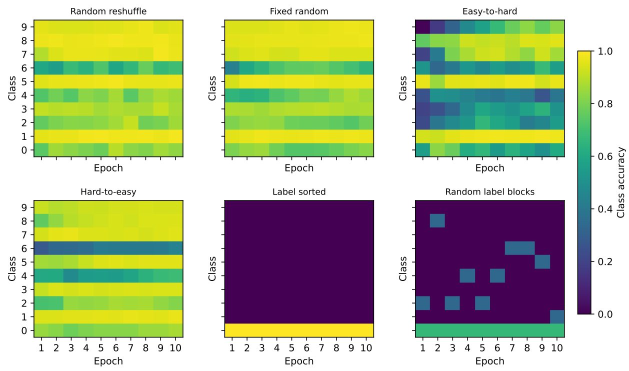

Final test accuracy, mean $\pm$ std across seeds.

## Appendix B: Gradient Cosine Similarity

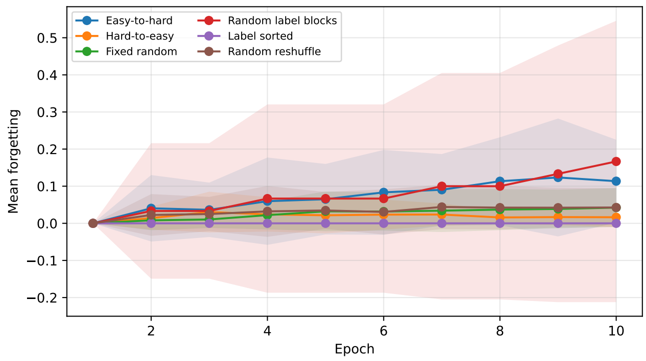



Successive gradient cosine similarity $\cos(\mathbf{g}_t, \mathbf{g}_{t-1})$ over optimization steps. High values indicate persistent directional alignment.

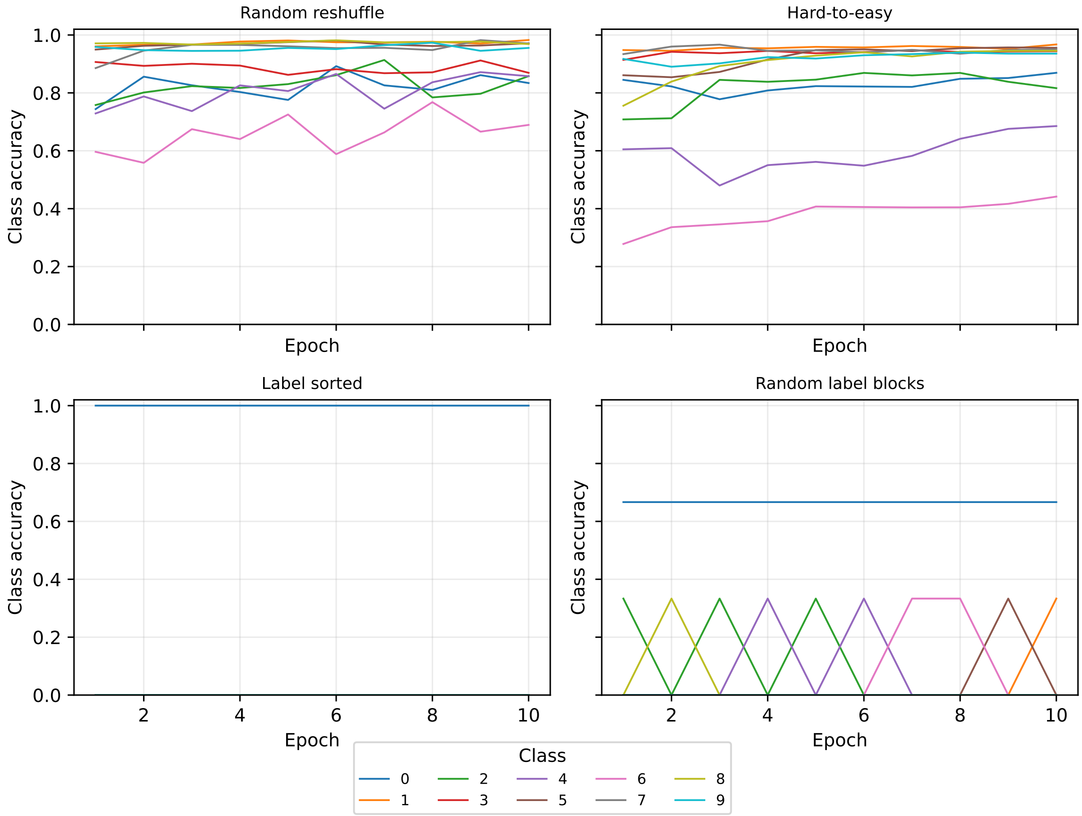
## Appendix C: Class Accuracy and Forgetting




Per-class test accuracy heatmaps (class × epoch).

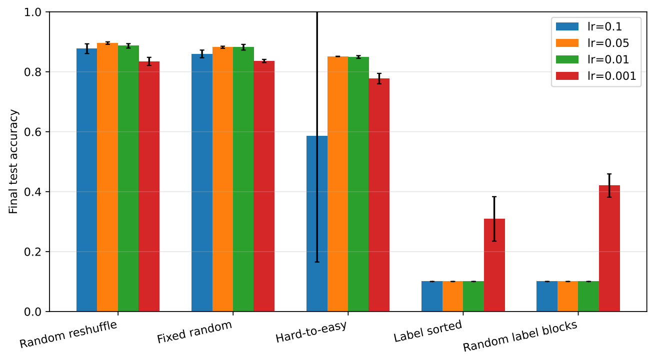



Mean forgetting $\bar{F}(t) = \frac{1}{C}\sum_c F_c(t)$ across classes and seeds.

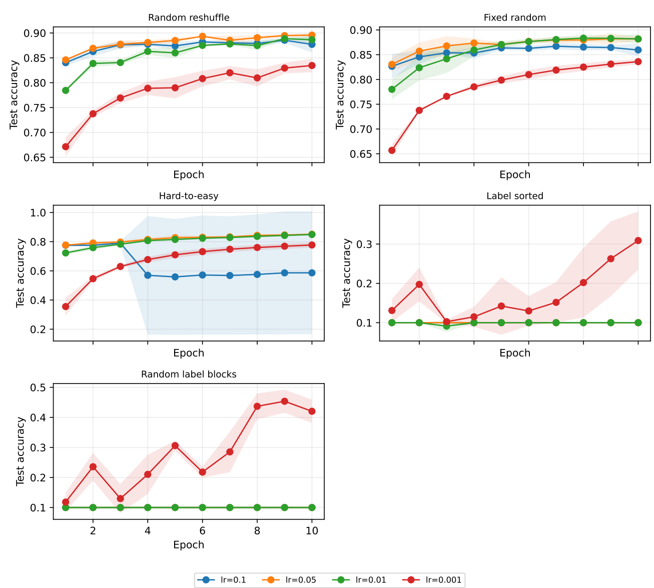



Per-class accuracy trajectories for four representative orderings.

# Appendix D: Learning-Rate Sensitivity

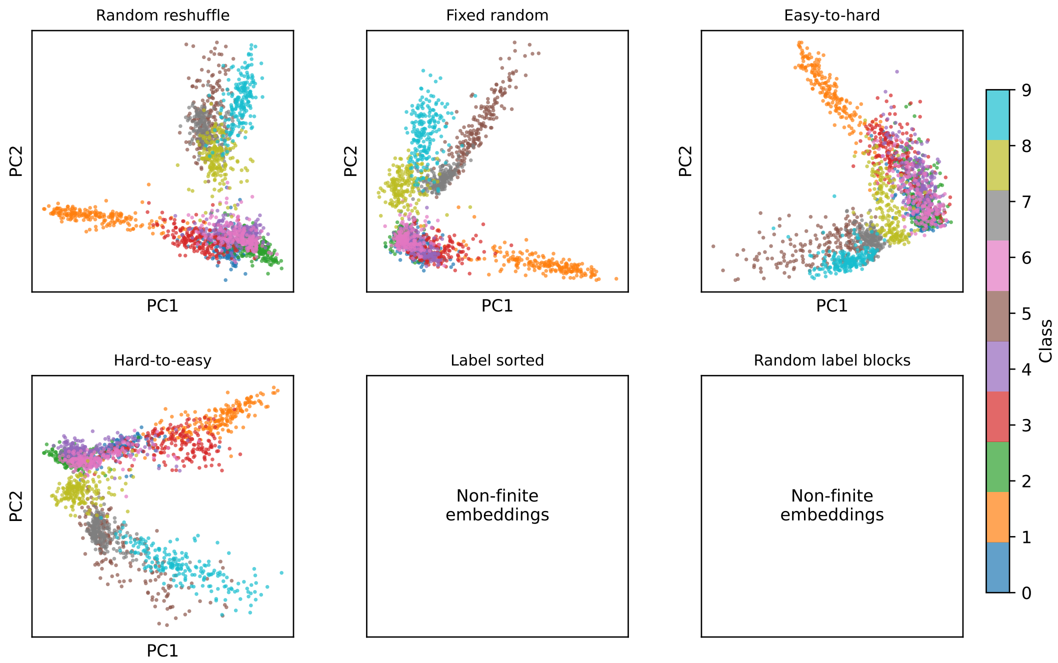



Final accuracy across $\eta \in \{0.001, 0.01, 0.05, 0.1\}$.



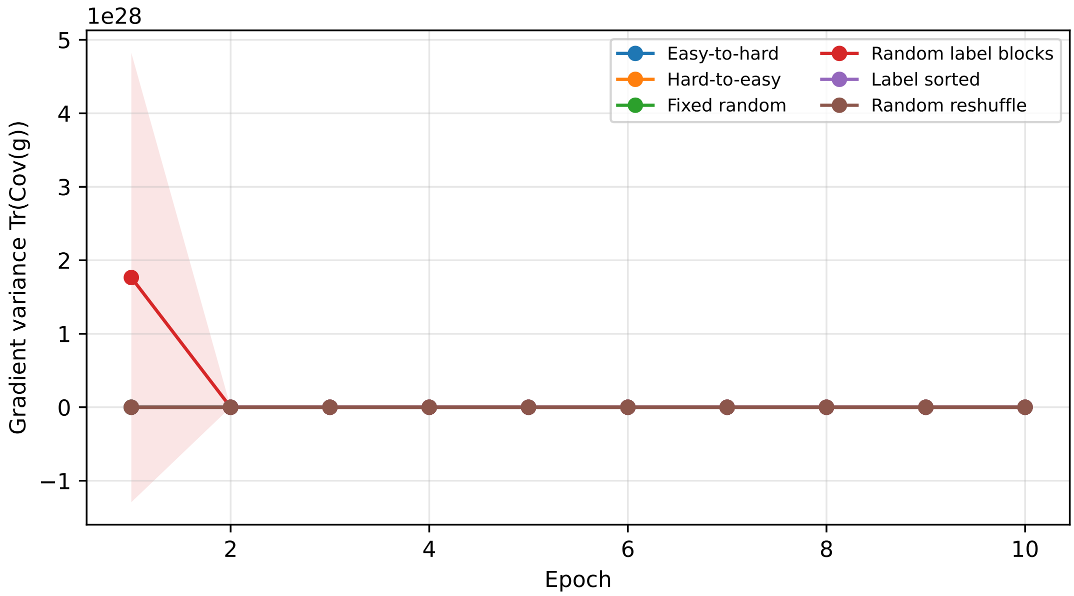

Accuracy trajectories for each ordering and learning rate.

## Appendix E: PCA Projections



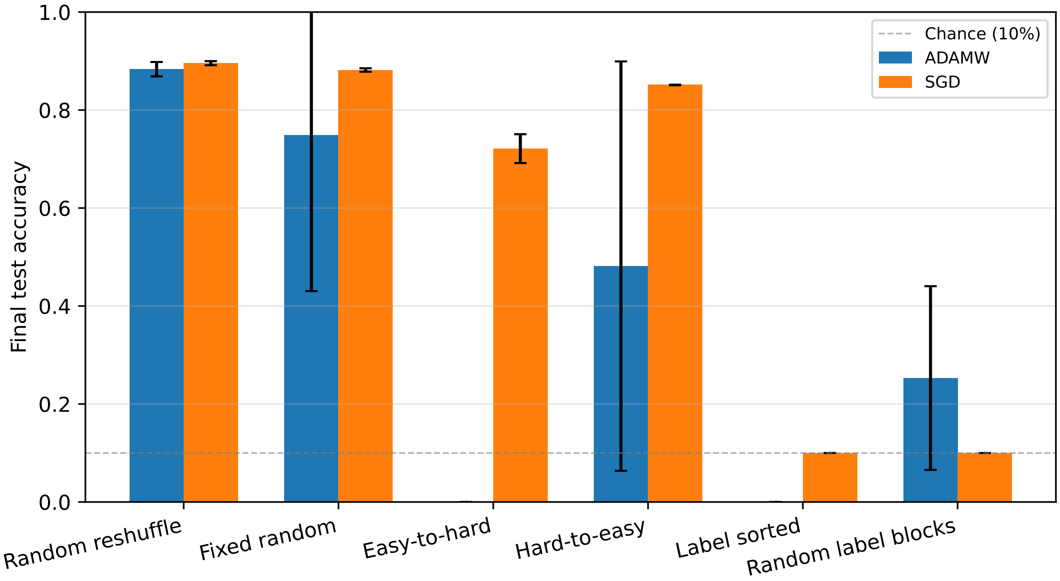

PCA of final hidden-layer embeddings on the test set (seed 0). Non-finite activations are labeled; they indicate a completely degenerate representation.

## Appendix F: Gradient Variance



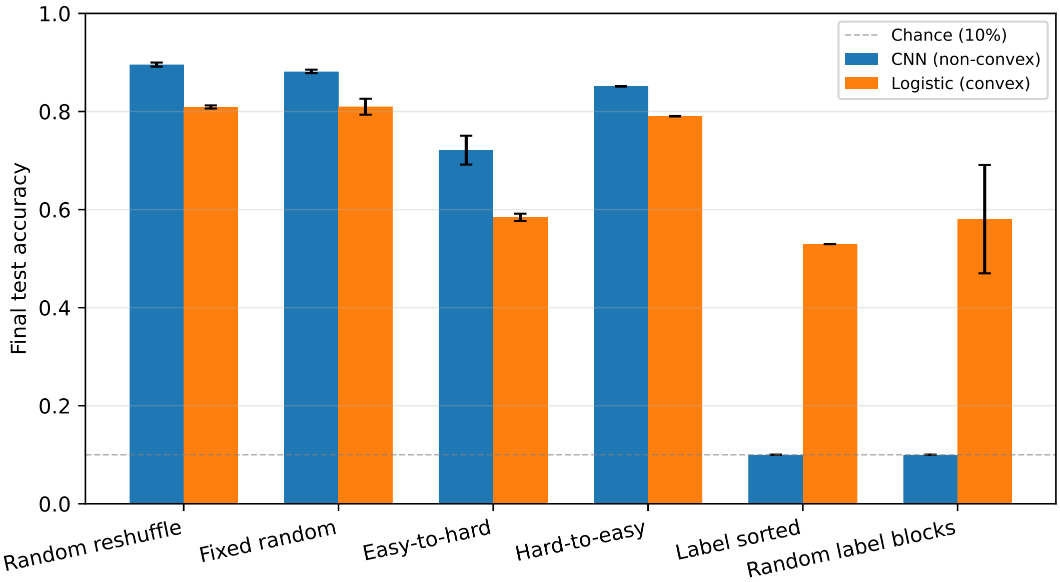

Per-epoch gradient variance $\sigma^2 = \mathrm{Tr}(\mathrm{Cov}(\mathbf{g})) = \mathbb{E}[\|\mathbf{g}\|^2] - \|\mathbb{E}[\mathbf{g}]\|^2$ for each ordering (mean $\pm$ std across seeds). Note that class-ordered streams exhibit *lower* variance than random reshuffling — the failure mode is high bias, not high noise.

## Appendix G: AdamW vs. SGD Comparison




Final test accuracy for SGD and AdamW across orderings. AdamW partially recovers under `label_block_random` but does not match random reshuffling, confirming that the root cause is directional bias rather than step-size mismatch.

## Appendix H: Convex Baseline (Logistic Regression)




CNN (non-convex) vs. logistic regression (convex) final test accuracy for all orderings. `label_sorted` collapses in both models; `label_block_random` is worse for the CNN, suggesting that non-convexity amplifies but does not create the ordering failure.

## References


Bengio, Yoshua, Jérôme Louradour, Ronan Collobert, and Jason Weston. 2009. "Curriculum Learning." *Proceedings of the 26th International Conference on Machine Learning*, 41–48.

Bottou, Léon, Frank E. Curtis, and Jorge Nocedal. 2018. "Optimization Methods for Large-Scale Machine Learning." *SIAM Review* 60 (2): 223–311.

Gürbüzbalaban, Mert, Asuman Ozdaglar, and Pablo A. Parrilo. 2021. "Why Random Reshuffling Beats Stochastic Gradient Descent." *Mathematical Programming* 186: 49–84.

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McCloskey, Michael, and Neal J. Cohen. 1989. "Catastrophic Interference in Connectionist Networks: The Sequential Learning Problem." *Psychology of Learning and Motivation* 24: 109–65.

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